DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL

FINAL EXAMINATION-2022

Course Title: Artificial Intelligence

Course Code: CSE-3205

3rd Year 2nd Semester

Session: 2019-20

Time: 3 hours **Marks:** 60

Answer any five Questions from the followings.

1. a) How is Machine Learning related to Artificial Intelligence? [3]

   b) Define Agent and Rational Agent through real-time example [3]

   c) Distinguish between the following properties of a task environment: [3]
      i. Static or dynamic
      ii. Discrete or continuous
      iii. Single or multi-agent

   d) What is local maxima? How to escape local maxima? [3]

2. a) Describe the function of vacuum cleaner as an Agent with suitable diagram. [4]

   b) Imagine, Mary is a model-based AI that builds and maintains an internal representation of the world. And Greg, on the other hand, is a goal-based AI that focuses on achieving specific objectives. [4]
   Describe the fundamental differences between Model-Based Mary and Goal-Based Greg in terms of their operational strategies. How does each approach affect their decision-making processes and adaptability to changes in their environment

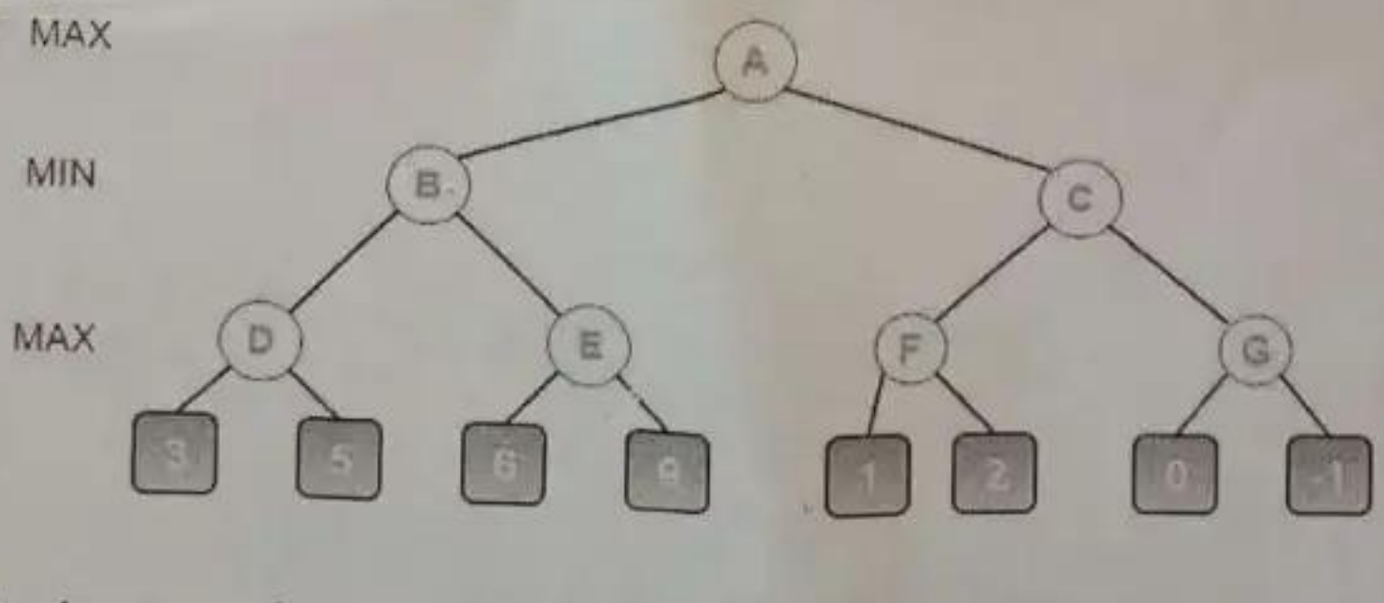
   c) Define Rational Agent with example. Is vacuum cleaner agent Rational? Why or why not? Explain with suitable reasons. [4]

3. a) Define in your own words the following terms: state, state space, search tree, search node, goal, action, transition model, and branching factor. [4]

   b) What is Greedy Best First Search? Explain with an example the different stages of Greedy Best First search. [4]

   c) Explain iterative deepening search with example. [4]

4. a) Explain the process of Minimax without alpha-beta pruning and how it determines the optimal move. [6]



   b) How does alpha-beta pruning improve the efficiency of the Minimax algorithm in evaluating game trees like the tree? [4]

   c) Explain the terms "max node" and "min node" in the Minimax algorithm. How are they used to represent players in a game? [2]

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